Class 23 Global Bifurcations

Teams 3 and 4: Post screenshots of your work to the course Google Drive today. Include words, labels, and other short notes that might make those solutions useful to you or your classmates. Find the link in Canvas.

Questions

1. For each of the following bifurcations, answer the following:

   - What is the minimum number of fixed points you expect in the phase portrait?
   - What kind of fixed point(s) could there be?
   - Does the number or type of fixed points change during the bifurcation?
   - How will the number of stable, and of unstable, limit cycles change during the bifurcation?

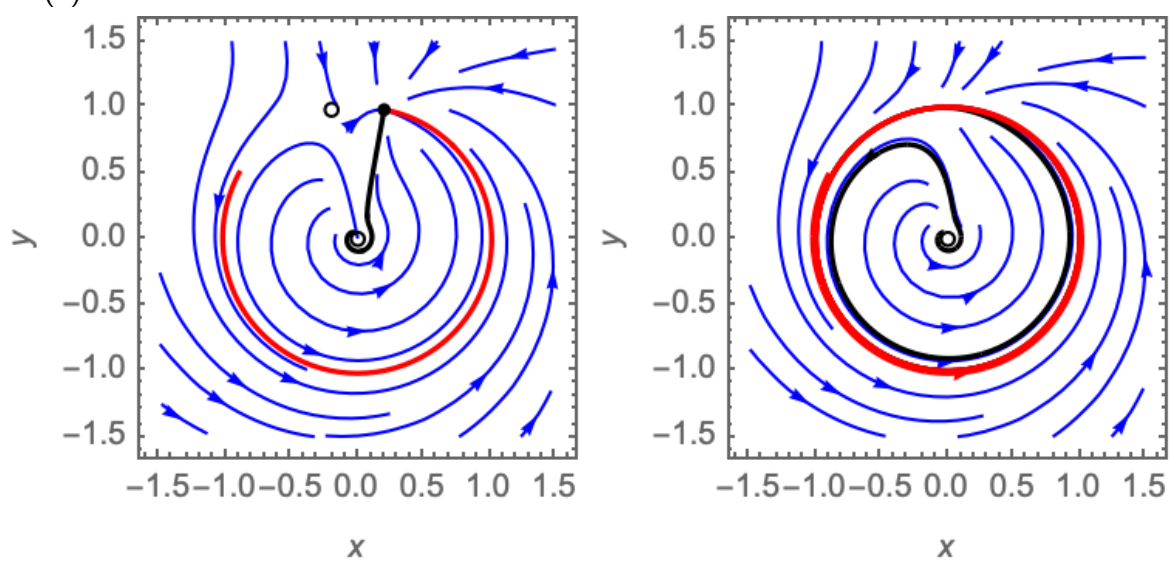
   (a) saddle-node of limit cycles

   (b) infinite-period

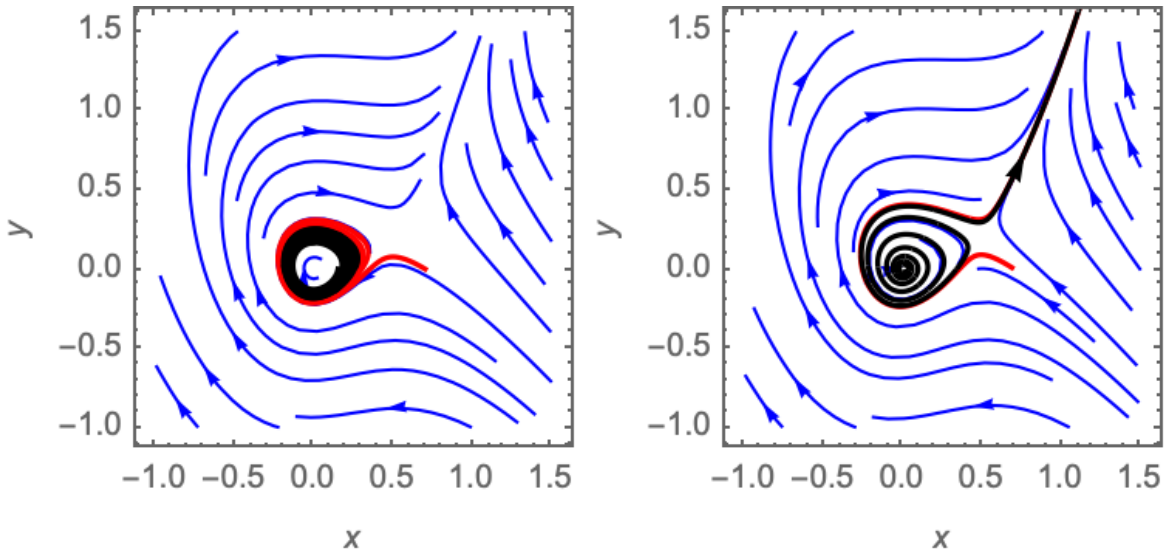
   (c) homoclinic

2. Each pair of phase portraits below are for a system on either side of a bifurcation. For each pair of phase portraits, fill in the chart and identify the bifurcation.

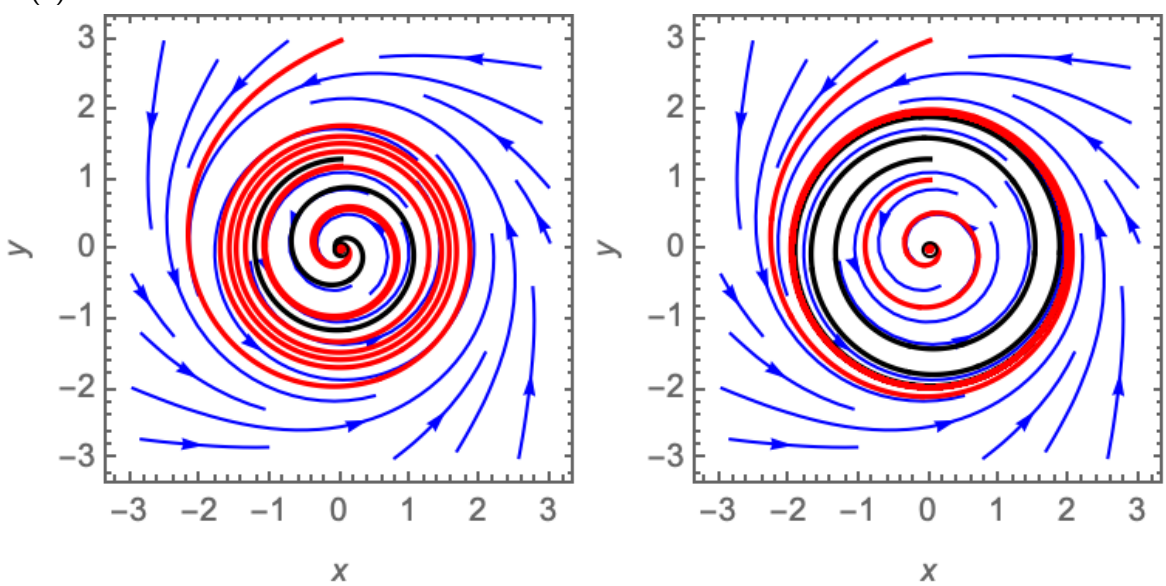
   (a) Bifurcation A



   (b) Bifurcation B



(c) Bifurcation C



	left	right
# of fixed points		
# of stable limit cycles		
# of unstable limit cycles		

Bifurcation:

◯ saddle node of cycles ◯ saddle node infinite period ◯ homoclinic

3. (Timing of bifurcations)

   (a) We have encountered the infinite-period bifurcation before, when we studied flows on the circle. In Class 10 we computed how long it takes to flow around the circle as we approached the bifurcation. We found that the time was proportional to $\frac{1}{\sqrt{r}}$, where r was the distance from the bifurcation in parameter space.

How is the bifurcation we studied in Class 10 related to the infinite-period bifurcation we are seeing in 2D?

(b) In a homoclinic bifurcation, a limit cycle approaches a saddle point. As the limit cycle becomes close to the saddle point, the time it takes to traverse the cycle is dominated by the time spent near the saddle point. Why is that?

(c) We can calculate the time spent near the saddle point using linearization. Let μ be the distance of the limit cycle from the stable manifold of the saddle point (measured along the unstable direction), at a distance k up the stable manifold from the saddle point.

Assume that the limit cycle is close enough to the saddle point that a linear approximation is appropriate to describe the dynamics around the saddle point.

In that case, solutions are $c_1 e^{\lambda_1 t}\mathbf{v}_1 + c_2 e^{\lambda_2 t}\mathbf{v}_2$ with $\lambda_1 < 0$ and $\lambda_2 > 0$. Let $\mathbf{v}_1$ and $\mathbf{v}_2$ each be length 1.

Choose c_1 and c_2 to match the scenario described. *One will be set to μ and the other to k.*

(d) A particle starting at the point above will flow past the saddle point to k distance along the unstable manifold when $c_i e^{\lambda_i T} = k$ (where you have determined whether $i = 1$ or $i = 2$ in your work above). Find an expression for T, the time when this occurs. How does this time depend on μ?

4. **1D map cobweb example**. $x_{n+1} = f(x_n)$ with $f(x) = 2x(1-x)$.

From left to right:

1. Start the orbit (and cobweb) at x_0 on the horizontal axis.
2. Draw a vertical line connecting x_0 to $x_1 = f(x_0)$. Find x_1 along the horizontal axis by drawing a horizontal line (x_0, x_1) to (x_1, x_1).

   x_0 and x_1 are marked along the horizontal axis.

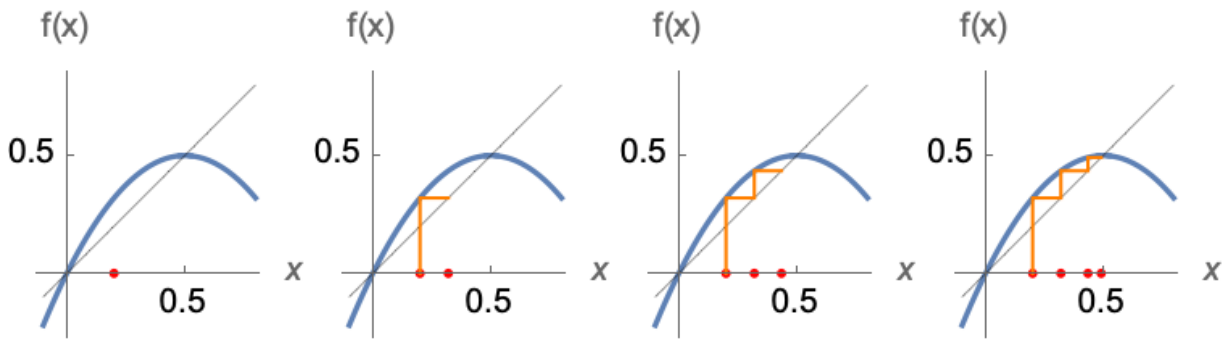
   The orbit lives in 1d. The pair (x_0, x_1) represents two consecutive points in the orbit.
3. Draw a vertical line from x_1 up to $x_2 = f(x_1)$. Find x_2 along the horizontal axis by drawing a horizontal line (x_1, x_2) to (x_2, x_2).

   x_0, x_1, x_2 are marked along the horizontal axis.

   The orbit lives in 1d. The pair (x_1, x_2) represents two consecutive points in the orbit.
4. Draw a vertical line from x_2 up to $x_3 = f(x_2)$. Find x_3 along the horizontal axis by drawing a horizontal line (x_2, x_3) to (x_3, x_3).

   x_0, x_1, x_2, x_3 are marked along the horizontal axis.

   The orbit lives in 1d. The pair (x_2, x_3) represents two consecutive points in the orbit.



Practice.

Let $x_{n+1} = f(x_n)$ with $f(x) = \cos x$.

(a) Plot $\cos x$ and x on the same plot.

(b) Use your plot to identify any fixed points of the map.

(c) Make cobwebs for at least two qualitatively different cases.